

Task 1: Objectives and Sub-Objectives of the Minor Project

Project Title: GuardShield AI: Real-Time LLM Guardrail Proxy

Main Objective

To develop a real-time AI guardrail proxy that protects Large Language Models (LLMs) by detecting and preventing prompt injection attacks and identifying hallucinations with minimal latency, providing a secure and reliable AI inference pipeline.

Sub-Objectives

Develop a Prompt Injection Detection Module

- Fine-tune a DeBERTa-v3 classifier to detect jailbreaks, DAN prompts, prompt leakage, and adversarial inputs.
- Block malicious prompts before they reach the LLM (Pre-Scan).

Implement Secure LLM Inference

- Route only validated prompts to a local LLM (Ollama Llama-3.1/Qwen-2.5).
- Ensure secure and efficient inference.

Develop a Hallucination Detection Module

- Monitor streaming output using token logit entropy.
- Verify generated content against reference context using NLI entailment scoring.
- Detect factual inconsistencies in real time (Post-Scan).

Evaluate Model Performance

- Benchmark using JailbreakBench, HackAPrompt, and HaluEval datasets.
- Measure Precision, Recall, F1-score, ROC-AUC, hallucination detection accuracy, and latency overhead.

Maintain Low-Latency Processing

- Keep additional proxy latency below 25 ms for real-time deployment.

Build a Working Minor Project Prototype

- Integrate pre-scan, LLM inference, and post-scan into a functional pipeline.
- Demonstrate the complete workflow with benchmark results.

Lay the Foundation for the Major Project

- Prepare the architecture for future FastAPI deployment, telemetry dashboard, autonomous red-teaming, and research publication.

Expected Outcome

A working GuardShield AI prototype capable of detecting prompt injections before inference and identifying hallucinations during streaming generation, supported by benchmark evaluation and designed for extension into a production-grade guardrail system.